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Q.1 A. Write a program that asks the user how many days are in a particular month, and

what day of the week the month begins on (0 for Monday, 1 for Tuesday, etc), and then

prints a calendar for that month. For example, here is the output for a 30-day month that begins on day 3 (Thursday):

M T W T F S S

- - - - - 1 2 3 4

5 6 7 8 9 10 11

In [28]:
```
# list = ['M','T','W','T','F','S','S']

days = int(input("Enter Number of Days in Month"))
start = int(input("Enter starting day of Month [0 to 6] : "))
print(" M  T  W  T  F  S  S")

print("   " * start, end="")

for day in range(1, days + 1):

    print(f"{day:2d}", end=" ")

    if (start + day) % 7 == 0:
        print()

print()
```

```
 M  T  W  T  F  S  S
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29
```

Q. 2. Check if all letters in a string are uppercase

In [29]:
```
s = "PYTHON FOR BEGINNERS"

flag = True

for char in s:
    if char == " ":
        continue

    if ord(char) < 65 or ord(char) > 90:
        flag = False
        break
if flag:
    print("Letters are in uppercase")
else:
    print("Letters are not in uppercase")
```

```
Letters are in uppercase
```

Q. 3. Write a version of a palindrome recognizer that also accepts phrase palindromes such as : Was it a rat I saw? or Dammit, I'm mad! Note that punctuation, capitalization, and spacing are usually ignored.

In [30]:
```
s1 = "Was it a rat I saw ?"
s2 = "Dammit, I'm mad!"
s3 = "Adesh"
def isPalindrome(s):
    print(s)
    result = ""
    for char in s:
        if char.isalnum():
            result += char.lower()
    if result == result[::-1]:
        print("It's a Palindrome\n")
    else:
        print("It's not a Palindrome\n")
isPalindrome(s1)
isPalindrome(s2)
isPalindrome(s3)
```

```
Was it a rat I saw ?
It's a Palindrome

Dammit, I'm mad!
It's a Palindrome

Adesh
It's not a Palindrome
```

Q.4. Write a Python function that takes a list of words and returns the length of the longest one

In [31]:
```
words = input("Enter items separated by space: ").split()
print(words)

maximum = 0

for word in words:
    if len(word)>maximum:
        maximum = len(word)
print(maximum)
```

```
['I', 'AM', 'IRONMAN']
7
```

Q.5. Write a Python program to remove duplicates from a list

In [32]:
```
l1 = ["red","yellow","white","grey","black","blue","white","yellow","grey"]
l2 = [1,2,3,4,5,7,4,5,7,8,9,3,]
# s1 = set(l1)
# l1 = list(s1)
# l1
def remove_duplicates(l):
    new = []
    for i in l:
        if i not in new:
            new.append(i)
    return new

print(remove_duplicates(l1))
print(remove_duplicates(l2))
```

```
['red', 'yellow', 'white', 'grey', 'black', 'blue']
[1, 2, 3, 4, 5, 7, 8, 9]
```

Q 6. Create a list of books

e.g. booklist =[['Java 8', 700], ['Python for Beginners', 500],....]

Perform following operations on the list

1. Add a new book with price
2. Remove entry for a book
3. update price for a book
4. Sort the list by book names
5. Sort the list by prices
6. Print the book with max and min price [hint : you may use min()/max() functions of python]

In [20]:
```
booklist = [['Java 8', 700],
            ['Python for Beginners', 500],
            ["Data Science using Pyhton", 1500]]
booklist
```

Out[20]:
```
[['Java 8', 700],
 ['Python for Beginners', 500],
 ['Data Science using Pyhton', 1500]]
```

In [21]:
```
#1. Add a new book with price
booklist.append( ['Artificial Intelligence A Modern Approach', 1200])
booklist
```

Out[21]:
```
[['Java 8', 700],
 ['Python for Beginners', 500],
 ['Data Science using Pyhton', 1500],
 ['Artificial Intelligence A Modern Approach', 1200]]
```

In [22]:
```
#2. remove entry for a book
booklist.pop(0)
```

Out[22]:
```
['Java 8', 700]
```

In [23]:
```
#3. update price for a book
booklist[0][1] = 1000
booklist
```

Out[23]:
```
[['Python for Beginners', 1000],
 ['Data Science using Pyhton', 1500],
 ['Artificial Intelligence A Modern Approach', 1200]]
```

In [24]:
```
#4. Sort the list by book names
booklist.sort()
booklist
```

Out[24]:
```
[['Artificial Intelligence A Modern Approach', 1200],
 ['Data Science using Pyhton', 1500],
 ['Python for Beginners', 1000]]
```

In [25]:
```
#Sort the list by prices
sort_by_price = sorted(booklist,key = lambda book:book[1])
sort_by_price
```

Out[25]:
```
[['Python for Beginners', 1000],
 ['Artificial Intelligence A Modern Approach', 1200],
 ['Data Science using Pyhton', 1500]]
```

In [26]:
```
#Print the book with max and min price [hint : you may use min()/max() functions
min_price = min(booklist, key=lambda book:book[1])
print(f'book with min price : {min_price}')
```

```
book with min price : ['Python for Beginners', 1000]
```

In [27]:
```
max_price = max(booklist, key=lambda book:book[1])
print(f'book with max price : {max_price}')
```

```
book with max price : ['Data Science using Pyhton', 1500]
```

Q.7. Write a Python program to compute element-wise sum of given tuples, using “zip()” function

Original tuples:

(1, 2, 3, 4)

(3, 5, 2, 1)

(2, 2, 3, 1)

Element-wise sum of the said tuples:

(6, 9, 8, 6)

In [18]:
```
t1 = (1,2,3,4)
t2 = (3,5,2,1)
t3 = (2,2,3,1)

zipped = tuple(x+y+z for x,y,z in zip(t1,t2,t3))
print(f'Element-wise sum : {zipped}')
```

```
Element-wise sum : (6, 9, 8, 6)
```

Q.8 Write a Python program to find the repeated items of a tuple.

In [16]:
```
t1 = (1,2,3,4,5,7,4,5,7,8,9,3)
repeated = tuple(set(x for x in t1 if t1.count(x)>1))
print(f'Repeated items : {repeated}')
```

```
Repeated items : (3, 4, 5, 7)
```

In [ ]:
```

```